

PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

Machine Learning-Based Cyber Attack Detection Using Security Event Data

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Machine Learning-Based Cyber Attack Detection Using Security Event Data**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Gifty Gupta**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

Date: 13/05/2026

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

The rapid growth of digital technologies and internet-based services has significantly increased the number and complexity of cyber threats across various sectors. Traditional cybersecurity systems mainly rely on rule-based and signature-based detection techniques, which are effective only for identifying known threats. However, these approaches often fail to detect new and evolving cyber attacks, creating a need for more intelligent and adaptive security solutions. Machine Learning has emerged as an effective approach for analyzing large volumes of security event data, identifying suspicious activities, and improving automated threat detection.

This study focuses on the implementation of a Machine Learning-based cyber attack detection system using structured cybersecurity event data. The main objective of the research is to analyze cybersecurity logs and classify different types of cyber attacks using data-driven techniques. The dataset used in this work contains information related to attack types, attack severity, threat intelligence, response actions, and other network-related attributes. Before model development, the dataset was preprocessed by removing irrelevant features, handling missing values, and converting categorical attributes into numerical form using label encoding techniques.

Two Machine Learning classification algorithms, Logistic Regression and Random Forest, were implemented and evaluated in this study. Logistic Regression was used as a baseline model due to its simplicity, while Random Forest was selected for its ability to handle complex and non-linear relationships within the dataset. The dataset was divided into training and testing sets using an 80:20 ratio, and model performance was evaluated using standard metrics such as accuracy, precision, recall, and F1-score. Various visualization techniques, including attack distribution graphs, correlation heatmaps, confusion matrices, and feature importance analysis, were also used to better understand the dataset and model behavior.

The experimental results indicate that both models faced challenges in accurately classifying cyber attacks due to weak feature relationships and limited predictive information within the dataset. Logistic Regression achieved an accuracy of approximately 19%, while Random Forest achieved around 20% accuracy, slightly outperforming the baseline model. The analysis highlights the importance of data quality, feature engineering, and preprocessing in improving Machine Learning performance for cybersecurity applications.

Although the achieved accuracy is relatively low, the study successfully demonstrates the practical implementation of Machine Learning techniques for cyber attack detection and security data analysis. The research emphasizes the potential of data-driven approaches in enhancing automated cybersecurity systems and reducing dependency on manual monitoring. Future improvements may include the use of advanced Deep Learning models, richer datasets, and real-time threat detection frameworks to improve overall performance and reliability.

1. Introduction

The rapid advancement of digital technologies and the increasing dependence on internet-based services have significantly transformed modern communication, business operations, healthcare, education, and government infrastructure. While these technological developments provide numerous benefits, they have also increased the risk of cyber threats and security attacks. Organizations today generate and store large volumes of sensitive information, making them attractive targets for cybercriminals. As a result, cyber attacks such as malware, phishing, ransomware, denial-of-service attacks, and unauthorized access attempts have become more frequent and sophisticated.

Traditional cybersecurity systems mainly rely on rule-based mechanisms and signature-based detection techniques to identify malicious activities. These approaches are effective for detecting previously known threats but often fail to recognize new, evolving, or complex attack patterns. Modern cyber attacks continuously change their behavior to bypass conventional security systems, making manual monitoring and static detection methods insufficient for current cybersecurity requirements. In addition, the massive amount of security event data generated by networks and systems makes manual analysis time-consuming and prone to human error.

To overcome these limitations, Machine Learning has emerged as a powerful solution for improving cyber attack detection and automated threat analysis. Machine Learning algorithms can analyze large-scale security datasets, identify hidden patterns, and classify malicious activities with minimal human intervention. Unlike traditional systems, Machine Learning models can learn from historical data and improve their performance over time, making them suitable for dynamic and unpredictable cybersecurity environments.

This research focuses on the implementation of a Machine Learning-based cyber attack detection system using structured cybersecurity event data. The study aims to analyze security logs, preprocess the dataset, identify important features, and apply Machine Learning classification techniques for detecting different types of cyber attacks. Two Machine Learning algorithms, Logistic Regression and Random Forest, are implemented and evaluated using standard performance metrics such as accuracy, precision, recall, and F1-score.

The study also emphasizes the importance of data preprocessing, feature selection, and visualization techniques in improving model performance and interpretability. Various graphical analyses such as attack distribution charts, correlation heatmaps, confusion matrices, and feature importance analysis are used to better understand the dataset and prediction behavior.

The main objective of this research is to develop an efficient and scalable data-driven approach for cyber attack detection that can support automated cybersecurity systems. Although the study highlights certain limitations related to dataset quality and feature relationships, it successfully demonstrates the practical application of Machine Learning techniques in cybersecurity data analysis and threat classification.

2. Methodology

The proposed methodology follows a systematic approach for detecting and classifying cyber attacks using Machine Learning techniques. The process includes data collection, preprocessing, feature selection, model training, and performance evaluation.

2.1. Data Collection

A structured cybersecurity event dataset was used in this study. The dataset contains information related to:

- Attack type
- Attack severity
- Threat intelligence
- Response actions
- Network-related attributes

The dataset was loaded using Python libraries such as Pandas for inspection and analysis.

2.2. Data Preprocessing

Data preprocessing was performed to improve dataset quality and prepare it for Machine Learning algorithms.

Steps Involved:

- Removal of missing and incomplete records

- Elimination of irrelevant attributes
- Cleaning noisy data

Features Removed:

- Event ID
- Timestamp
- Source IP Address
- Destination IP Address
- User Agent

These features were removed because they did not contribute significantly to attack prediction.

2.3. Data Encoding and Transformation

The dataset contained categorical attributes that required conversion into numerical format.

Technique Used:

- Label Encoding

This transformation ensured compatibility with Machine Learning algorithms, which require numerical input data.

2.4. Exploratory Data Analysis (EDA)

EDA was performed to understand dataset characteristics and identify important patterns.

Visualizations Used:

- Attack type distribution graph
- Attack severity distribution graph
- Correlation heatmap

The analysis revealed weak feature correlations, which affected overall prediction performance.

2.5. Feature Selection

Feature selection was carried out to retain only the most relevant attributes for model training.

Objectives:

- Reduce dimensionality
- Improve computational efficiency
- Enhance model interpretability

The selected features were used as input variables, while the attack type was considered the target variable.

2.6. Data Splitting

The dataset was divided into:

- 80% Training Data
- 20% Testing Data

The training dataset was used for model learning, while the testing dataset was used for performance evaluation on unseen data.

2.7. Machine Learning Models

Two classification algorithms were implemented in this study.

a) Logistic Regression

- Used as the baseline model
- Suitable for linear classification problems
- Simple and computationally efficient

b) Random Forest

- Ensemble-based classification algorithm
- Handles complex and non-linear relationships
- Reduces overfitting using multiple decision trees

Both models were trained using the processed dataset.

2.8. Model Evaluation

The performance of the models was evaluated using standard classification metrics.

Evaluation Metrics:

- Accuracy
- Precision
- Recall
- F1-Score

Additional Analysis:

- Confusion Matrix
- Accuracy Comparison Graph
- Feature Importance Analysis

These evaluation techniques helped analyze model behavior and prediction effectiveness.

2.9. Tools and Technologies Used

The implementation was carried out using the following tools and libraries:

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn

These technologies were used for data preprocessing, visualization, Machine Learning implementation, and performance analysis.

2.10. Overall Workflow

The proposed system follows a structured Machine Learning pipeline for cyber attack detection and classification. The workflow begins with dataset collection and preprocessing, followed by feature selection and data encoding. The processed dataset is then divided into training and testing sets for model development. Machine Learning models are trained and evaluated using standard performance metrics, and the final results are analyzed through graphical visualizations and performance analysis.

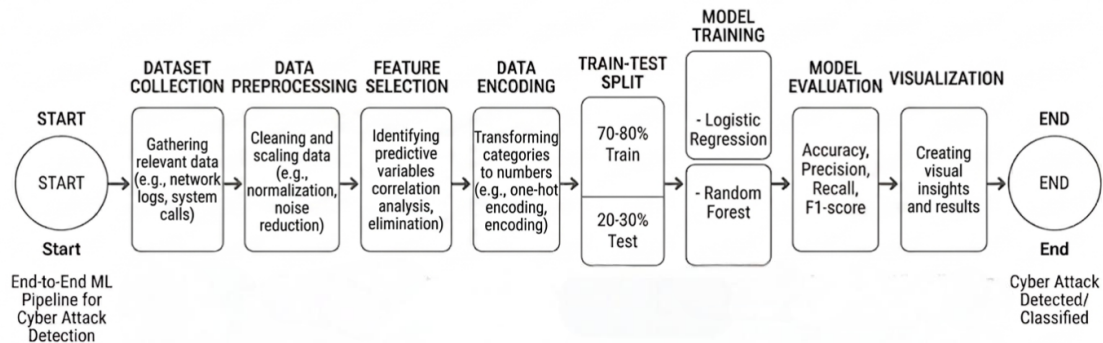


Fig. 1 Flowchart of Proposed Methodology

The overall workflow of the proposed methodology is illustrated in Fig. 1.

1. Data Collection
2. Data Preprocessing
3. Feature Selection
4. Data Encoding
5. Train-Test Splitting
6. Model Training
7. Model Evaluation

8. Visualization and Result Analysis

This methodology provides a systematic and practical approach for implementing Machine Learning-based cyber attack detection systems.

3. Tools and Technologies Used

The implementation of the proposed cyber attack detection system was carried out using Python and various Machine Learning and data analysis libraries. These tools were used for dataset preprocessing, model training, visualization, and performance evaluation.

3.1. Python

Python was used as the primary programming language due to its simplicity, flexibility, and extensive support for Machine Learning and data analysis applications.

3.2. Jupyter Notebook

Jupyter Notebook was used as the development environment for writing, executing, and testing the implementation code interactively.

3.3. Pandas

The Pandas library was used for:

- Data loading
- Data cleaning
- Dataset manipulation
- Handling structured cybersecurity event data

3.4. NumPy

NumPy was used for numerical computations and array-based operations required during data processing and model implementation.

3.5. Scikit-learn

Scikit-learn was used for implementing Machine Learning algorithms and evaluation techniques.

Functionalities Used:

- Logistic Regression
- Random Forest Classifier
- Label Encoding
- Train-Test Split
- Accuracy, Precision, Recall, and F1-Score Evaluation

3.6. Matplotlib

Matplotlib was used for generating graphical visualizations such as:

- Accuracy comparison graphs
- Attack distribution charts
- Performance visualizations

3.7. Seaborn

Seaborn was used for creating advanced statistical visualizations, including:

- Correlation heatmaps
- Confusion matrices
- Feature relationship analysis

These tools and technologies collectively helped in developing a structured and efficient Machine Learning-based cyber attack detection system.

4. Implementation

The implementation of the proposed cyber attack detection system was carried out using Python in a Jupyter Notebook environment. The system was developed using a structured workflow that includes data loading, preprocessing, Machine Learning model implementation, visualization, and performance evaluation.

4.1. Development Environment

The implementation was performed using Python because of its strong support for Machine Learning and data analysis applications. Jupyter Notebook was used as the development platform for executing code and visualizing outputs interactively.

Several Python libraries were used during implementation:

- Pandas for data handling and preprocessing
- NumPy for numerical operations
- Scikit-learn for Machine Learning models and evaluation
- Matplotlib and Seaborn for data visualization

4.2. Data Loading and Inspection

The cybersecurity event dataset was loaded using the Pandas library. Initial inspection was performed to understand:

- Dataset structure
- Data types
- Number of records
- Feature distribution

Functions such as `head()` and `info()` were used to examine the dataset and identify inconsistencies.

	Attack Type	Attack Severity	Data Exfiltrated	Response Action
0	Ransomware	Critical	False	Eradicated
1	Malware	Critical	True	Eradicated
2	Ransomware	High	False	Eradicated
3	DDoS	Critical	False	Contained
4	Malware	Medium	False	Eradicated

Fig. 2 Sample View of Processed Dataset

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20000 entries, 0 to 19999
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Event ID              20000 non-null  object
1   Timestamp             20000 non-null  object
2   Source IP             20000 non-null  object
3   Destination IP        20000 non-null  object
4   User Agent            20000 non-null  object
5   Attack Type           20000 non-null  object
6   Attack Severity       20000 non-null  object
7   Data Exfiltrated      20000 non-null  bool
8   Threat Intelligence   20000 non-null  object
9   Response Action       20000 non-null  object
dtypes: bool(1), object(9)
memory usage: 1.4+ MB

```

Fig. 3 Dataset Structure and Data Types

4.3.Data Preprocessing Implementation

Data preprocessing was carried out to improve dataset quality and prepare it for Machine Learning algorithms.

Preprocessing Steps:

- Removal of missing and incomplete records
- Elimination of irrelevant attributes
- Reduction of noisy data

Features Removed:

- Timestamp
- Source IP Address
- Destination IP Address
- User Agent
- Event ID

Categorical variables were converted into numerical form using label encoding techniques to ensure compatibility with Machine Learning algorithms.

4.4. Data Visualization

Visualization techniques were used to understand dataset characteristics and feature relationships.

Visualizations Performed:

- Attack type distribution graph
- Attack severity distribution graph
- Correlation heatmap

The visual analysis indicated weak feature correlations within the dataset, which affected model prediction performance.

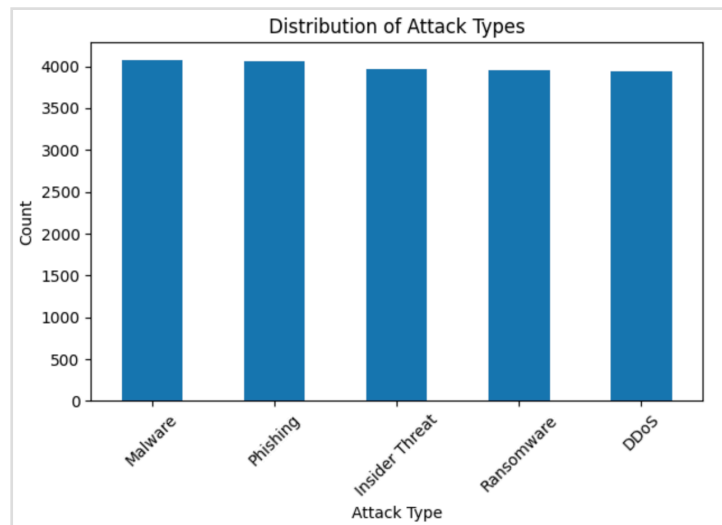


Fig. 4 Distribution of Cyber Attack Types

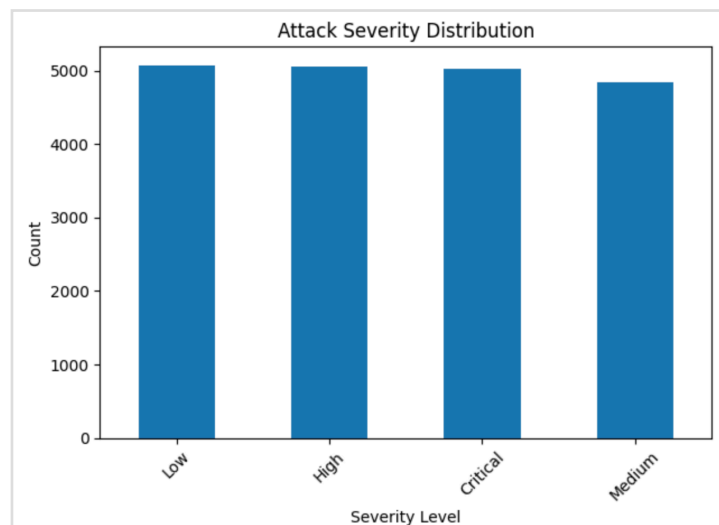


Fig. 5 Attack Severity Distribution

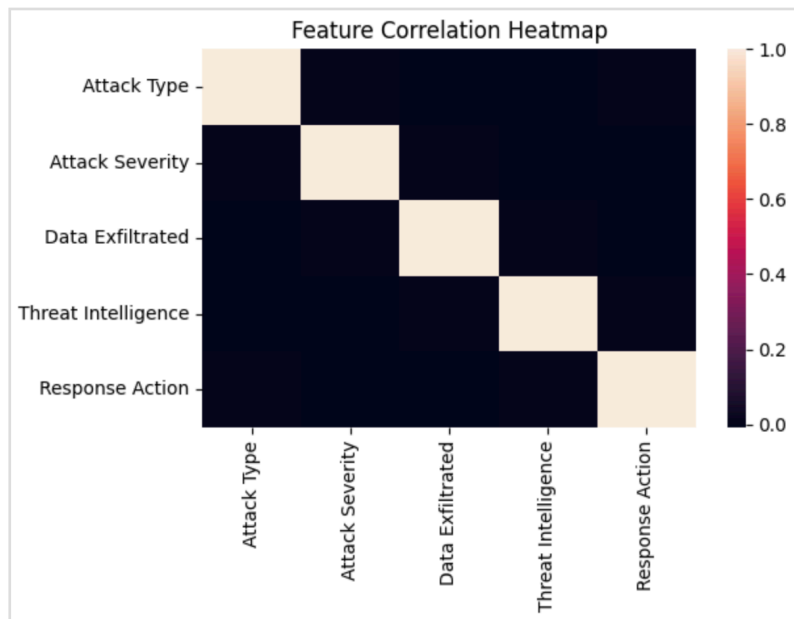


Fig. 6 Feature Correlation Heatmap

4.5. Model Implementation

Two Machine Learning classification algorithms were implemented:

a) Logistic Regression

- Used as the baseline model
- Suitable for linear classification tasks
- Simple and computationally efficient

b) Random Forest

- Ensemble-based classification model
- Handles complex and non-linear relationships
- Reduces overfitting using multiple decision trees

Both models were trained using the processed training dataset and tested on unseen data.

4.6. Model Evaluation Implementation

The performance of the models was evaluated using standard classification metrics.

Evaluation Metrics Used:

- Accuracy
- Precision
- Recall

- F1-Score

Additional analyses were also performed using:

- Confusion Matrix
- Accuracy Comparison Graph
- Feature Importance Analysis

These analyses helped evaluate model behavior and classification effectiveness.

5. Results

The performance of the implemented Machine Learning models was evaluated using standard classification metrics such as accuracy, precision, recall, and F1-score. Two classification algorithms, Logistic Regression and Random Forest, were applied to detect and classify cyber attack types using the processed cybersecurity dataset.

5.1. Model Performance Analysis

The experimental results indicate that both Machine Learning models achieved relatively low classification accuracy due to weak feature relationships within the dataset. Logistic Regression achieved an accuracy of approximately 19%, while the Random Forest model achieved a slightly higher accuracy of around 20%.

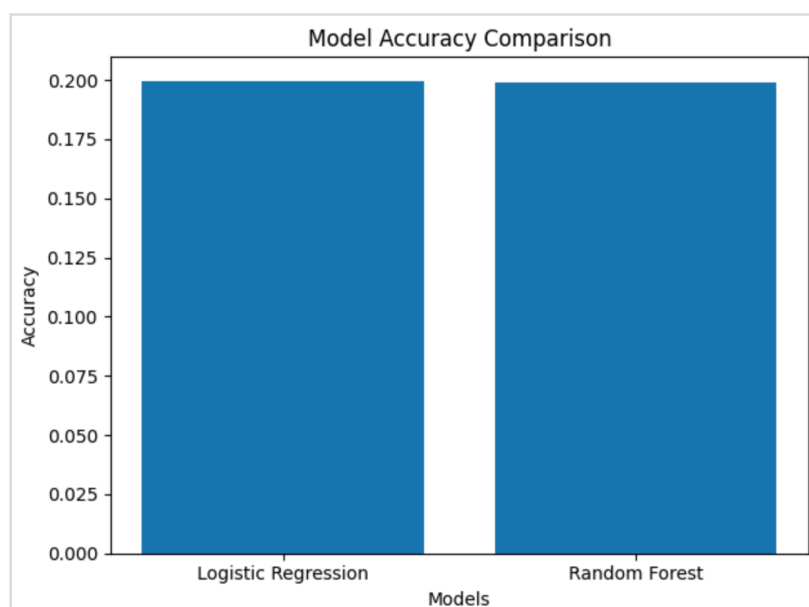


Fig. 7 Model Accuracy Comparison

Although the improvement in performance was marginal, Random Forest performed better because of its ability to capture non-linear relationships and reduce overfitting using multiple decision trees. However, the absence of strong predictive features limited the overall effectiveness of both models.

5.2. Confusion Matrix Analysis

The confusion matrix generated for the Random Forest model showed that predictions were distributed across multiple attack classes, with frequent misclassification among categories. The matrix did not show strong concentration along the diagonal, indicating that the model struggled to correctly distinguish between several attack types.

This behavior suggests that the dataset lacked clearly separable feature patterns, making classification more complex. The analysis also highlights the challenges faced by Machine Learning models when trained on datasets with weak correlations and limited discriminative information.

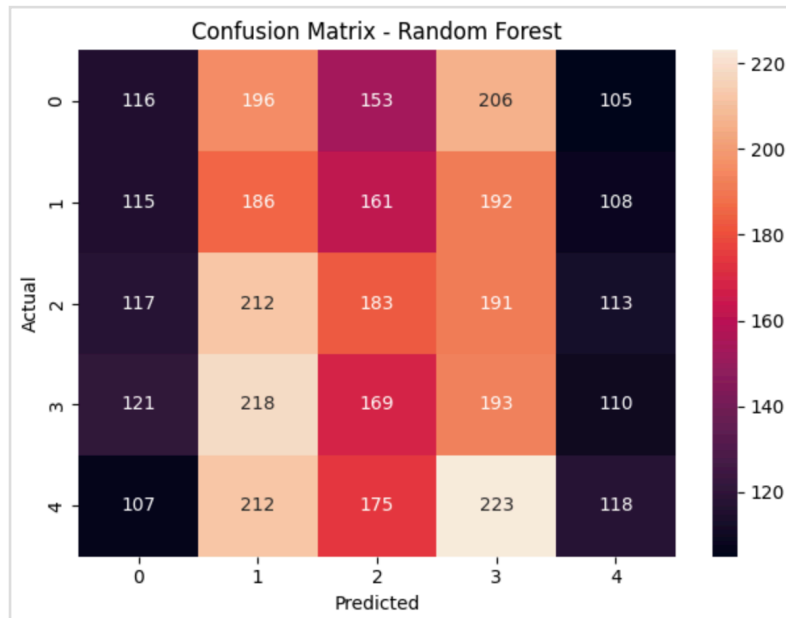


Fig. 8 Confusion Matrix of Random Forest Model

5.3. Feature Importance Analysis

Feature importance analysis was performed using the Random Forest model to identify the contribution of individual features toward cyber attack classification. The analysis revealed that attributes such as **threat intelligence** and **attack severity** contributed relatively more compared to other features.

However, no single feature showed a dominant influence on prediction accuracy. Most features contributed only moderately, indicating that the dataset lacked strong predictive attributes required for accurate classification. This finding emphasizes the importance of feature engineering and high-quality datasets in Machine Learning-based cybersecurity systems.

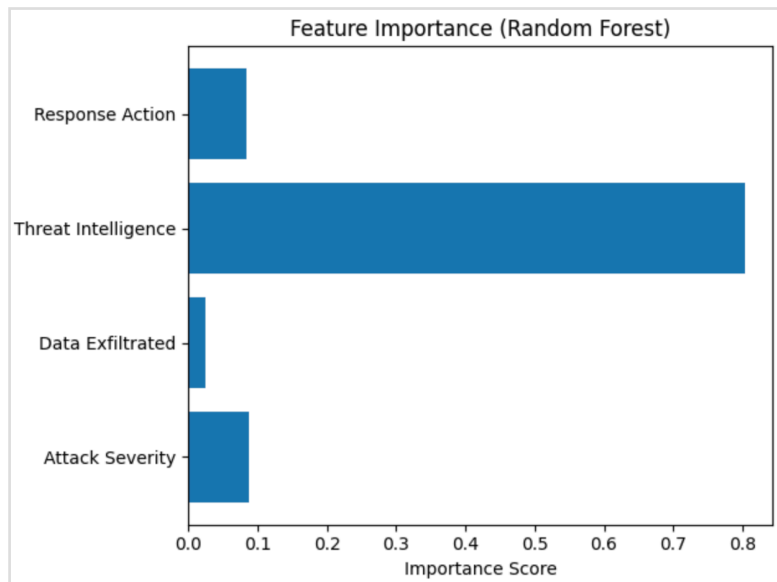


Fig. 9 Feature Importance Analysis

5.4. Overall Findings

The study demonstrates that Machine Learning techniques can be effectively implemented for cybersecurity data analysis and automated threat detection. However, the performance of the models is highly dependent on dataset quality, feature relevance, and the presence of meaningful patterns within the data.

The results also highlight the importance of proper preprocessing, feature selection, and evaluation techniques in improving model interpretability and prediction performance.

Future improvements may include advanced feature engineering methods, richer datasets, and the use of more sophisticated Machine Learning and Deep Learning models.

6. Conclusion and Future Scope

6.1. Conclusion

This study presented a Machine Learning-based approach for detecting and classifying cyber attacks using structured cybersecurity event data. The implementation involved multiple stages, including data preprocessing, feature selection, data encoding, model training, and performance evaluation. Two Machine Learning classification algorithms, Logistic Regression and Random Forest, were implemented and analyzed to identify different cyber attack patterns.

The experimental results showed that both models achieved relatively low classification accuracy, with Logistic Regression achieving approximately 19% accuracy and Random Forest achieving around 20% accuracy. Although Random Forest slightly outperformed Logistic Regression, the overall performance remained limited due to weak feature relationships and insufficient predictive information within the dataset.

The analysis demonstrated that dataset quality and feature relevance play a major role in determining Machine Learning performance in cybersecurity applications. The visualizations and evaluation results highlighted that the dataset lacked strong correlations among features, making it difficult for the models to distinguish effectively between different attack categories.

Despite these limitations, the study successfully demonstrated the practical implementation of Machine Learning techniques for cybersecurity data analysis and automated threat detection. The research also emphasized the importance of preprocessing, feature selection, and proper evaluation methods in developing reliable cyber attack detection systems.

Overall, the project provides a strong foundation for understanding the application of Machine Learning in cybersecurity and highlights the challenges involved in building effective attack classification models using real-world security event data.

6.2. Future Scope

The study can be extended and improved in several ways to enhance cyber attack detection performance and reliability.

Possible Future Improvements:

- Use of advanced Machine Learning and Deep Learning models such as:
 - Support Vector Machines (SVM)
 - Gradient Boosting
 - Neural Networks
 - Deep Learning architectures
- Implementation of advanced feature engineering techniques to extract more meaningful patterns from cybersecurity data.
- Use of richer and more informative real-world datasets containing stronger predictive relationships among features.
- Development of real-time cyber attack detection systems capable of continuously monitoring network traffic and responding dynamically to threats.
- Application of optimization and hyperparameter tuning techniques to improve model performance and prediction accuracy.
- Reduction of noise, redundant features, and class imbalance issues within the dataset.
- Integration of Machine Learning models into live cybersecurity frameworks for automated threat analysis and faster response systems.

Future research in these areas may significantly improve the efficiency, accuracy, and scalability of Machine Learning-based cybersecurity systems.

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